

Birla Institute of Technology and Science Pilani (Rajasthan)
Department of Electronics and Electrical Engineering
Analog Digital VLSI Design (EEE/INSTR F313)
First Semester 2016-17

Date : 05/12/2016

Time: 09:00 AM to 12:00 PM

COMPREHENSIVE EXAM
(Part-A Closed Book)

Max. Time: 55 minutes

Max.Marks:30

Solutions

Instructions to the candidate

Please write your answers legibly and neatly in the **separate answer Sheet provided** only.

Crossed Answers shall not be considered for a recheck.

There is **no partial marking** in this part of the exam.

All questions are compulsory and carry **3.0 marks** each. **No Negative marking.**

You may use the **supplementary answer sheet** for rough work which **will not be considered for evaluation.**

Common data for PART-A: Use the following data if not mentioned specifically in the question.

$V_{DD} = 3V$, $|V_{TOP}| = 0.8V$, $V_{TON} = 0.6V$, $KP_P = 50 \mu A/V^2$, $KP_N = 150 \mu A/V^2$, $C_L = 100fF$, $f = 850MHz$,
 $L_{min}(2\lambda) = 0.3\mu m$.

Each question carries **THREE** marks. Write your answers in the separate answer sheet provided.

1. Assuming minimum length transistors calculate the ration of width of PMOS and NMOS in a static CMOS inverter that produces an output voltage of 2.2 V for an input voltage of 1.2V.

Solution:

For the given conditions the PMOS is in linear region of operation while the NMOS is in saturation region of operation. So equating the currents through them

The terminal voltages for PMOS are $V_{SD} = 3.0 - 2.2$, $V_{SG} = 3.0 - 1.2$

So for PMOS ($0.8 < 1.8-0.8$) is in Linear region

The terminal voltages for NMOS are $V_{GS} = 1.2 - 0$, $V_{DS} = 2.2 - 0$

So NMOS ($2.2 > 0.6$) is in Saturation region

Equating the currents through PMOS and NMOS

$$50 \times W_P(2(1.8 - 0.8)0.8 - 0.8^2) = 150 \times W_N(1.2 - 0.6)^2$$

$$W_P = 1.125W_N$$

2. In static CMOS inverter if the rise time and fall time of the input signal increases, then which component of the power consumption changes and why? (Max. 3 sentences)

Solution: When the rise time/ fall time of the input signal increases the duration for which both the **PMOS and NMOS are simultaneously ON** increase as a result the **short circuit power** consumption increases. (P.No.221 DIC Rabey 2e)

3. Justify the statement “The switching power consumption at the output node of a pass transistor logic is low in comparison to static CMOS family”. (Max. 3 sentences)

Solution: In pass transistor logic family the **output voltage swing is reduced**, since the output node can't reach rail to rail (0 to $V_{DD} - V_{TN}$ for NMOS and $|V_{TP}|$ to V_{DD} in PMOS) as a result the switching power reduces. (P.No.271 DIC Rabey 2e)

4. If a design engineer wants to selectively eliminate the body effect for PMOS in a given circuit, what exactly does he/she mean from the fabrication perspective. (Max. 3 sentences)

Solution: Yes, it is possible to eliminate body effect in PMOS by tying the source of each PMOS to its body. This is possible in n-well process where each PMOS has its own well. This comes at the expense of increased area since the minimum well to well spacing rules apply.

5. Is it possible to reduce the delay of a critical path, hence the overall circuit delay by modifying the threshold voltage of the MOSFETs on the critical path? If so, how? (Max. 3 sentences)

Solution: Yes, it is possible to do so by reducing the threshold voltage of the MOSFETs on the critical path this increases the drive current of these MOSFETs and hence the delay reduces. The penalty paid is an additional implant mask for such transistors.

6. For the arrangement shown in **fig.1** (Note: *cd-contamination delay*, *pd-propagation delay* and *su-setup time*). Calculate the maximum clock frequency if the positive skew 25ps.

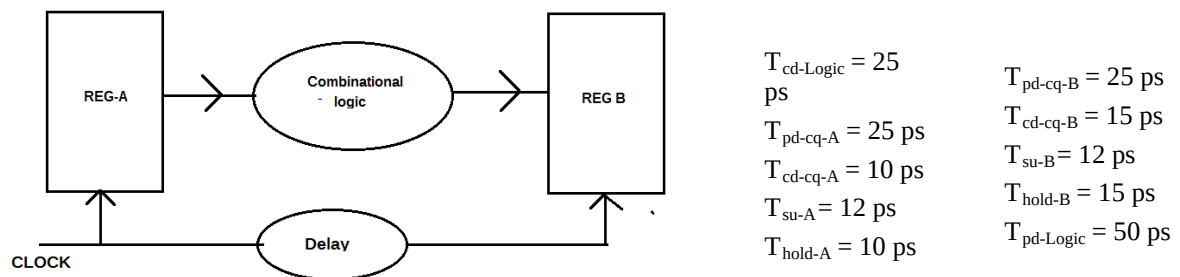


Figure 1: Question no.6

Solution:

$$T(\min) = T_{pd-cq}(\text{reg} - A) + T_{pd-logic} + T_{su}(\text{reg} - B) - T_{skew}$$

Substituting the values $T(\min) = 25 + 50 + 12 - 25 = 62\text{ps} \Rightarrow f_{\max} \sim 16\text{GHz}$

But look at the other condition $T_{hold}(\text{reg} - B) < T_{cd-logic} + T_{cd-cq}(\text{reg} - A) - T_{skew}$

$$15\text{ps} < 25\text{ps} + 10\text{ps} - 25\text{ps} \text{ is false}$$

So the circuit doesn't work faithfully for any clock frequency until the hold time violation is resolved by using additional buffers in the data path (P.No. 497 DIC Rabey 2e)

7. Match the following logic families in List-A with the properties listed in List-B. (report the appropriate alphabet in the space provided)

S.No.	List-A (logic family)			List-B (property)
1.	DCVSL	[_____]	[a]	Rail to Rail Swing
2.	Pseudo NMOS	[_____]	[b]	Zero Static Power Consumption
3.	Transmission Gate	[_____]	[c]	Ratioed logic
			[d]	Dynamic logic

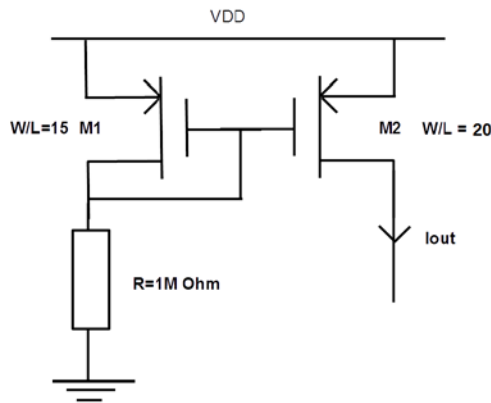
Solution:

DCVSL : Ratioed, Rail to Rail swing and Zero Static Power consumption. (a,b,c)

Pseudo NMOS : Ratioed Logic (b)

Transmission gate : Rail to Rail Swing and Zero Static Power consumption. (a,b)

8. Calculate the current I_{out} from **fig.2**. (neglect channel length modulation)



Solution

let, the voltage on the gate of PMOS be V

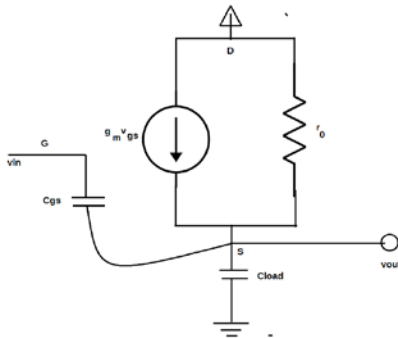
$$\frac{V}{R} = 0.5 \times 50 \times 15 \times (3 - V - 0.8)^2$$

$$V = 2.277V \text{ or } 2.124V$$

Only 2.124 V is the possible value else PMOS is in cut-off so the current is

$$2.124\mu A \Rightarrow I_{out} \text{ is } 2.832 \mu A$$

9. For the source follower circuit shown in **fig.3**. Write the expressions for pole/s and zero/s If any?



Solution (P.No. 179 B.Razavi)

There is a Miller capacitor in the form of C_{gs} so there will be a zero and a pole.

$$pole = -\left(\frac{g_m}{C_{gs} + C_{load}}\right)$$

$$zero = -\frac{g_m}{C_{gs}}$$

Fig.3. Equivalent circuit of source follower

10. Write an expression for positive and negative slew rate (C_1 and C_2 dominate over parasitic capacitances) for the single ended Op-Amp shown in **fig.4**.

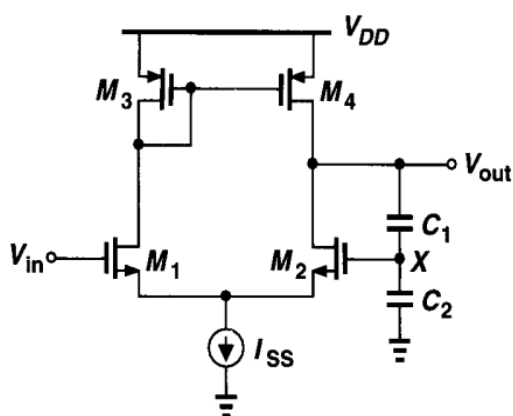


Fig.4. Question 10

Solution (P.No.331 B.Razavi)

The negative and positive slew rate are the same in magnitude

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$Positive Slew rate = \frac{I_{SS}}{C_{eq}}$$

$$negative Slew rate = -\frac{I_{SS}}{C_{eq}}$$

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Section B

Q.1

Given $AE(Q_1) : AE(Q_2) = 10:1$
where AE is emitter area of transistors

voltage at point b $\Rightarrow V_b = R_1 I_{PTAT} + V_{EB1}$

" " " a $\Rightarrow V_a = V_{EB2}$

The opamp will force $V_a = V_b$

$$I_{PTAT} \cdot R_1 = V_{EB2} - V_{EB1}$$

$$= V_T \ln \left(\frac{I_{C2}}{I_{C1}} \right) = V_T \ln \left(\frac{I_C}{10 I_C} \right)$$

$$I_{PTAT} \cdot R_1 = V_T \ln 10$$

$$I_{PTAT} = \frac{V_T \ln 10}{R_1}$$

The net current through $m_1 / m_2 / m_3$ is

$$I = I_{PTAT} + \frac{V_{EB2}}{R_2}$$

(3M)

hence o/p voltage reference

$$V_{REF} = R_3 \left(\frac{V_T \ln 10}{R_1} + \frac{V_{EB2}}{R_2} \right)$$

(5M)

(ii) In order to get zero temp. coeff. (neglect Temp. coeff. resistors)

$$\frac{\partial V_{REF}}{\partial T} = 0 \quad \frac{\partial V_T}{\partial T} \frac{\ln 10}{R_1} + \frac{\partial V_{EB2}}{\partial T} \frac{1}{R_2} = 0$$

$$0.087 \frac{mV}{K} \cdot \frac{\ln 10}{R_1} = \frac{1.91 \frac{mV}{K}}{R_2}$$

$$\frac{R_2}{R_1} = \frac{1.91}{0.087 \ln 10} = \frac{21.95}{\ln 10} = 9.535$$

$$[R_2 = 9.535 R_1]$$

(5M)

(iii) To realize $V_{REF} = 0.67V$

$$R_1 = 2K\Omega \text{ (given)}$$

$$R_2 = 19.07K\Omega \text{ (for ZTC)}$$

$$0.67 = \frac{R_3}{2} \left(\frac{26mV \times 2.30}{2} + \frac{0.7}{19.07} \right)$$

$$0.67 = R_3 (0.066)$$

$$R_3 = 10.05K\Omega$$

(5m)

Q2

32

Q. 2

Nitin

1) Fully differential cascode op-amp
or Telescopic op-amp.

$$\text{ICMR} \Rightarrow V_{in(\min)} = V_{OV(I_{SS})} + V_{gs} \quad \left(\begin{array}{l} V_{gs} = \\ V_{OV} + V_T \\ = 0.9 \end{array} \right)$$

$$= 0.2 + 0.7 + 0.2 = 1.1 \text{ V}$$

$$V_{in(\max)} = V_{DD} - 3V_{OV} + V_T = 3.3 \text{ V}$$

2) o/p signal swing

$$= 2(2.9 - 3(0.2))$$

$$= 2 \times 2.3 = 4.6 \text{ V}$$

3) P.D = 93 μA , $I_{bias} \approx 28 \mu\text{A}$

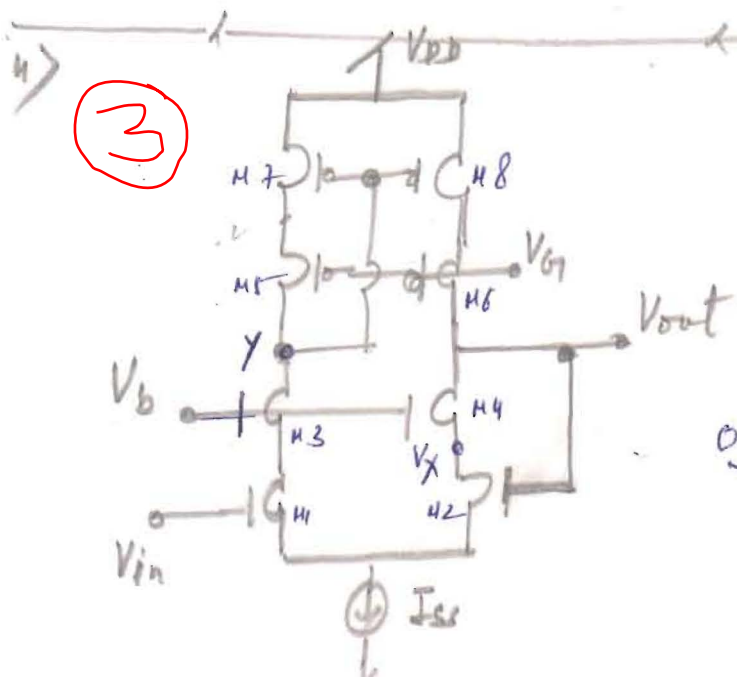
$$g_{m0n} = g_{m0p} = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 14 \mu\text{A}} \approx 714 \text{ K}\Omega$$

$$g_{m0n} = g_{m0p} = \frac{2I_D}{V_{OV}} = \frac{2 \times 14}{0.2} = 140 \mu\text{A/V}$$

$$A_v = \frac{(g_m g_o)^2}{2} = \frac{(140 \times 10^{-6} \times 714 \times 10^3)^2}{2}$$

$$A_v = 2498 = 67 \text{ dB}$$

- Slew rate = $\frac{I_{SS}}{C_L} = \frac{28 \mu\text{A}}{2 \text{ pF}} = 14 \text{ V}/\mu\text{s}$



$$V_b - V_{t4} \leq V_{out} \leq V_b - V_{gs4} + V_{t2}$$

OCMR $\Rightarrow$ Max $\rightarrow V_{DD} - 2V_{OV}$

$$= 2.9 \text{ V}$$

OCMR Min $\rightarrow V_{th4} - (V_{gs4} - V_{t2})$

$$= 0.7 - (0.2)$$

$$\approx 0.5 \text{ V}$$

$$V_{DD} = 3.3V \quad L = 1\mu m \quad \textcircled{B}$$

8.3. $C_{ox} = 0.04 \text{ pF}/\mu m^2$, $V_{ov} = 0.2V$, $I_{SS} = I_7 = 60\mu A$, $\lambda = 0.1 V^{-1}$

(i) Small Signal Voltage gain $A = V_{out}/V_{in}$

$$A \approx (g_{m2} r_{o5}) \times \left[g_{m7} (r_{o7} \parallel r_{o8}) \right] \left(\frac{1}{2} \right)$$

$$\left\{ \begin{aligned} g_{m2} &= \frac{2I_{SS/2}}{V_{ov}} = \frac{2 \times 30\mu A}{0.2} = \frac{300\mu A}{V}, r_{o5} = \frac{1}{0.1 \times 30\mu A} = 333 k\Omega \\ g_{m7} &= \frac{2I_7}{V_{ov}} = \frac{2 \times 60\mu A}{0.2} = \frac{600\mu A}{V}, r_{o7} = r_{o8} = \frac{1}{0.1 \times 60\mu A} = 166 k\Omega \end{aligned} \right.$$

$$A \approx (300 \times 333 \times 600 \times 166) \times 10^{-6} \times \frac{1}{2} \approx 10,000 / 2 = 5000$$

$\approx 74 \text{ dB}$ $\left(\frac{1}{2} \right)$

(ii) $C_{gs} = \frac{2}{3} W L C_{ox}$ (for MOS in saturation region)

$$C_{gs1} = C_{gs2} = \frac{2}{3} \times 1\mu m \times 0.04 \text{ pF}/\mu m^2 \times W = 0.027 \frac{\text{pF}}{\mu m} W_{1,2}$$

$$W_1 = W_2 = W_3 = W_4 = \frac{g_{m2} \times L}{\mu_n C_{ox} V_{ov}} = \frac{300 \times 1 \frac{\mu A}{V} \times \mu m}{140 \frac{\mu A}{V} \times 0.2 V} = 10.7 \mu m$$

$$W_5 = W_6 = \frac{300 \times 1}{40 \times 0.2} = \frac{75 \mu m}{2} = 37.5 \mu m$$

$$W_7 = W_8 = \frac{600 \times 1}{40 \times 0.2} = 75 \mu m$$

$$W_8 = \frac{600 \times 1}{140 \times 0.2} = 21.4 \mu m$$

$$C_{gs1} = C_{gs2} = 0.027 \text{ pF} \times 10.7 = C_{gs3} = C_{gs4} = 0.289 \text{ pF}$$

$$C_{gs5} = C_{gs6} = 0.027 \text{ pF} \times 37.5 = 1.01 \text{ pF}$$

$$C_{gs7} = 2.025 \text{ pF}$$

$$C_{gs8} = 0.578 \text{ pF}$$

0.3

(iii) Pole frequencies $\rightarrow$

① $\omega_{x,y} \approx -\frac{1}{(r_{o3} \parallel g_{m3} r_{o4}) C_{xy}}$

$C_{xy} \approx C_{gs3,9} + C_{gd4} + C_{gd6} + C_{db6} + C_{db4}$ (negligible)

① $\omega_{a,b} \approx -\frac{g_{m3}}{2 C_a}$; $C_{a,b} \approx C_{gs3,4} + C_{gd1} + C_{db3}$ (negligible)

① $\omega_{out2,1} \approx -\frac{1}{(r_{o7} \parallel r_{o8}) C_L}$

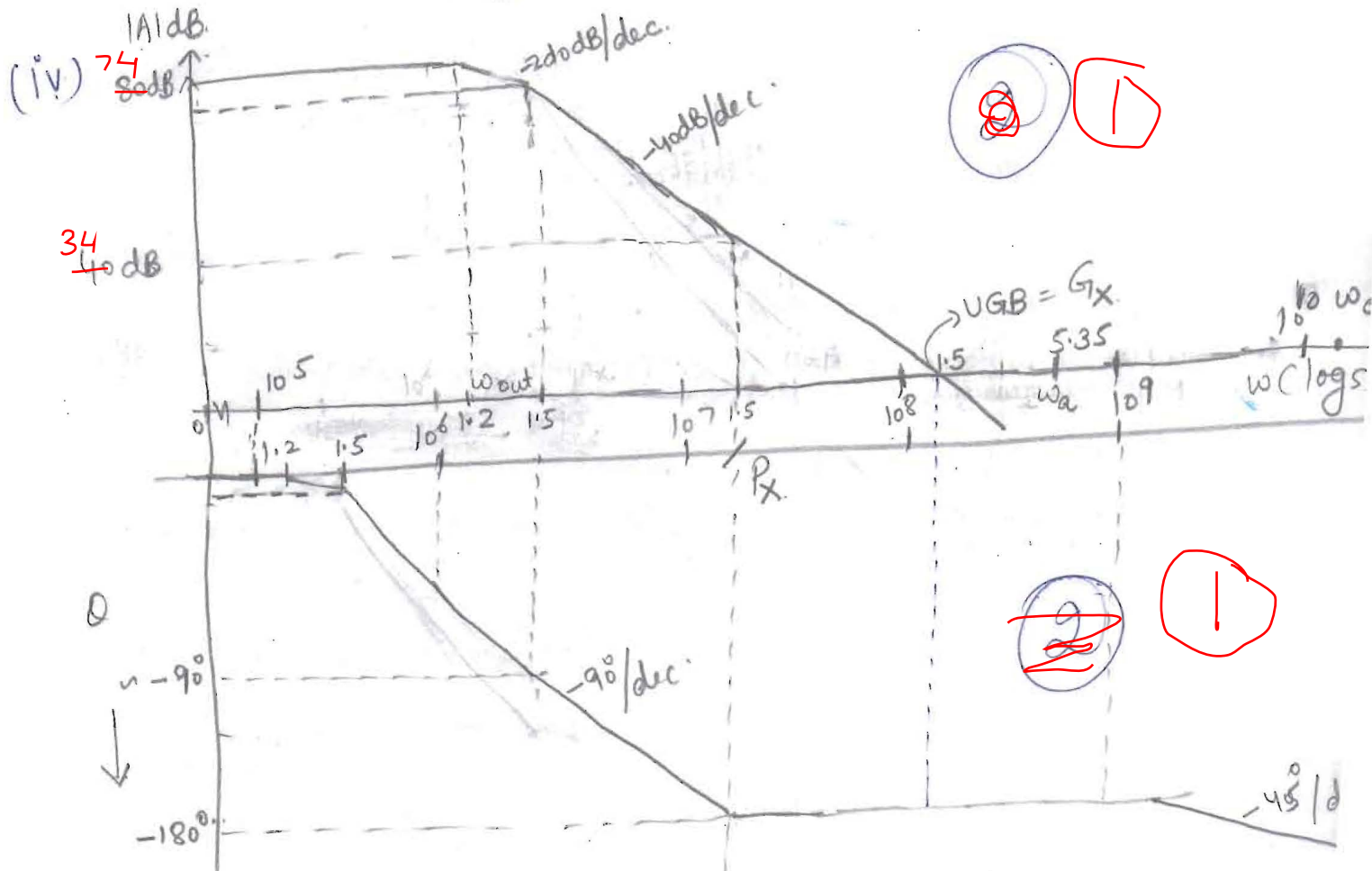
$\omega_{xy} \approx \frac{1}{333 K \Omega \times 2 p} \approx 1.5 \times 10^6 \text{ rad/sec}$; $\omega_{ab} \approx \frac{3000 \mu A}{2 \times 0.28 p} \approx 5.35 \times 10^8 \text{ rad/sec}$; $\omega_{out2} \approx \frac{1}{83 K \Omega \times 0 p} \approx 1.2 \times 10^6 \text{ rad/sec}$

④ Zero frequencies $\rightarrow$

2 feed forward path; through C_{gd1} , C_c .

① $\omega_{Cgd1} \approx \frac{g_{m2}}{C_{gd2}} \approx \frac{300 \mu}{0.028 p} \approx 1.07 \times 10^{10} \text{ rad/sec}$

$\omega_{Cc} \approx \frac{g_{m7}}{C_c + C_{gd1}} \approx \frac{600 \mu}{0.7 pF} \approx 857.1 \times 10^6 = 8.57 \times 10^8 \text{ rad/sec}$



(V) phase Margin $\approx 0^\circ$ (^{little} more than 0°)] ①
System is unstable.

(VI) To obtain a phase margin of 45° (approx.), Total phase $\approx -135^\circ$
Second pole ω_{xy} location should be at $\omega_{xy} = \omega_{GB}$.
or

$$\Rightarrow -\tan^{-1} \frac{\omega}{\omega_{p1}} - \tan^{-1} \frac{\omega}{\omega_{p2}} = -135^\circ$$

$$\omega = \omega_{GB} = \omega_{p2} = \omega_{xy} = 1.5 \times 10^6 \text{ rad/sec.}$$

$$\Rightarrow -\tan^{-1} \frac{1.5 \times 10^6}{\omega_{p1}} = -90^\circ \quad (\text{less } 6^\circ)$$

$$\Rightarrow \omega_{p1} \approx 1.57 \times 10^5 \text{ rad/sec.} = \omega_{out2}$$

To obtain $\omega_{out2} \approx 1.57 \times 10^5 \text{ rad/sec.}$ ①

Value of $C_c \Rightarrow$

$$\omega_{out2} = \frac{1}{(r_{o1} \parallel r_{o2}) C_{L_{new}}} = \frac{1}{83k C_{L_{new}}} = 1.57 \times 10^5 \text{ rad/sec.}$$

$$C_{L_{new}} = \frac{1}{\omega_{out2} \times 83k} = \frac{1}{1.57 \times 10^5 \times 83k} = \frac{8.032 \times 10^{-9}}{10+8} = 8.03 \times 10^{-11} \approx 80.3p$$

$$C_{L_{new}} = C_L + A_2 C_c = 10pF + 100 C_c = 80p$$

$$\Rightarrow \textcircled{1} C_c = \frac{70p}{100} = 0.7pF$$

Hence, Zero frequency will shift to $\omega_{Z_{gd}} = \frac{g_{m7}}{C_c} \approx \frac{600\mu}{0.7p} = 85.7 \times 10^6 \text{ rad/sec.}$

Q1 Ha

$$V_x = V_{DD} - V_T$$

$$\begin{aligned} \textcircled{a} \quad V_x &= V_{DD} - [V_{T0} + \gamma(\sqrt{V_{SB} + |2\phi_F|} - \sqrt{-2\phi_F})] \\ &= 2.5 - [0.43 + 0.5(\sqrt{V_x + 0.6} - \sqrt{0.6})] \end{aligned}$$

$$V_x = 2.5 - [0.43 - 0.39 + 0.5\sqrt{V_x + 0.6}]$$

$$V_x = 2.46 - 0.5\sqrt{V_x + 0.6}$$

$$\left(\frac{V_x - 2.46}{0.5}\right)^2 = -0.5\sqrt{V_x + 0.6}$$

$$V_x = 1.70V$$

2

②

$$V_x = V_{DD} - V_T$$

$$V_{DS2} = V_{DD} - V_{out}$$

$$V_{GS2} = V_T = V_{DD} - V_T - V_{out} - V_T = V_{DD} - V_{out} - 2V_T$$

As $V_{DS2} > V_{GS2} - V_T$, so M2 = saturation region.

2

③

$I_N = 0$, then $M_1 = \text{off}$

$$V_{out}(\text{Max}) = V_x - V_T$$

$$= V_{DD} - V_T - V_T$$

$$= V_{DD} - 2V_T$$

$$= 2.5 - 2 \times 0.43$$

$$= 1.65V$$

2

of b

(a)

$$V_{OL} = 0.4V, \text{ and } V_{OH} = 2.07V$$

$$V_{OL} = |V_{EP0}| = 0.4V$$

$$L = V_{DD} - V_{tn0} = 2.7 - 0.43 = 2.07V$$

2

(b)

$$I_D = 0.5 \mu n C_{ox} \left(\frac{W}{L} \right) (V_{gs} - V_t)^2$$

$$I_{D-0.4V} = 0.823 \text{ mA}$$

$$I_{D-1.24V} = 0.29 \text{ mA}$$

$$I_{avg} = 0.56 \text{ mA}$$

1

2

$$t_{PLH} = \frac{C_L \Delta V}{I_{avg}} = \frac{(100 \text{ fF}) (1.24V - 0.4V)}{0.56 \text{ mA}} = \frac{84 \text{ fF}}{0.56 \text{ mA}} = 150 \times 10^{-12} \text{ s} = 150 \text{ psec}$$

(c)

$$E_{supply} = C V_{DD} (V_{DD} - V_{tn0} - |V_{tp0}|)$$

$$= 100 \text{ fF} \times 2.5 (2.5 - 0.43 - 0.4)$$

$$= 250 \text{ fF} (2.07 - 0.4)$$

$$= 250 \text{ fF} (1.67)$$

$$E_{supply} = 417.5 \times 10^{-10} \text{ J}$$

1

1

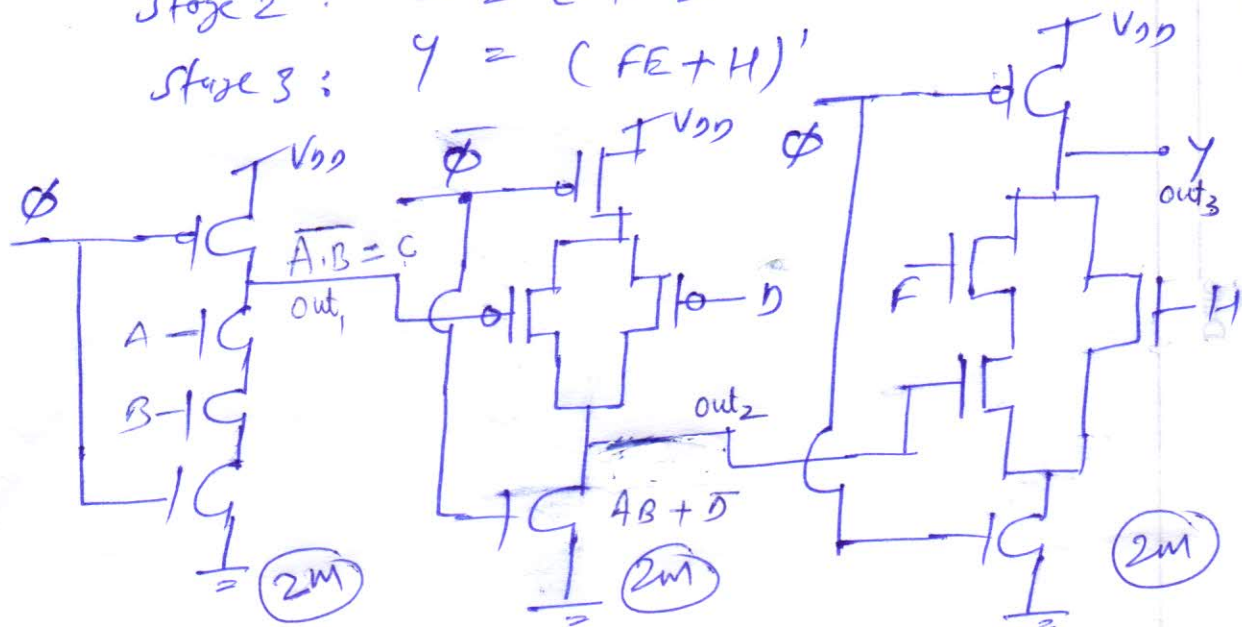
Q5

Q:

Ans: (i) Stage 1: $C = (A \cdot B)'$

Stage 2: $E = C' + D'$

Stage 3: $Y = (FE + H)'$



(ii) A, B, E & H must do transition from 0 to 1 only.
D must do transition from 1 to 0 only. (2M)

(iii) $T = T_{\text{precharge/discharge}} + 3T_{\text{evaluate}}$

$$= 2 + 3 \times 10 = 32 \text{ ns}$$

$$f_{\text{max}} = 1/T = 31.25 \text{ MHz}$$

For stage 1 & stage 3:

$$\begin{aligned} \Delta Q_{\text{critical, min}} &= C_{x, \text{min}} (V_{DD} - V_{IH}) \\ &= 10 \times 10^{-15} (5 - 2.9) \\ &= 21 \text{ fC} \end{aligned}$$

$$t_{\text{hold}_3} = t_{\text{hold}_1} = \frac{\Delta Q_{\text{critical, min}}}{I_{\text{leakage, max}}} = 0.0215 = 21 \text{ ms.}$$

For stage 2

$$\begin{aligned} \Delta Q_{\text{critical, min}} &= C_{x, \text{min}} (V_{IL}) = 10 \times 10^{-15} (1.9) \\ &= 19 \text{ fC} \end{aligned}$$

$$t_{\text{hold}_2} = \frac{\Delta Q_{\text{critical, min}}}{I_{\text{leakage, max}}} = 0.0195 = 19 \text{ ms.}$$

$$\Rightarrow f_{\text{min}} = \frac{1}{t_{\text{hold}_2}} = 52.63 \text{ Hz}$$